AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

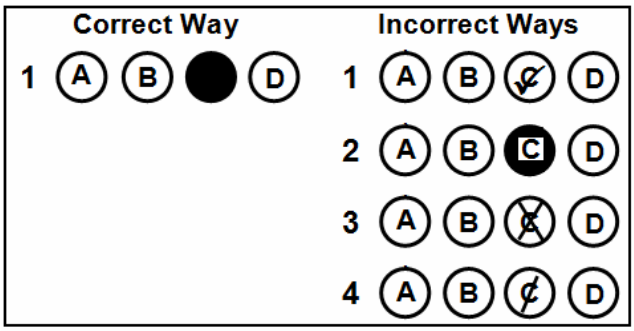
MAY 2013

Biology Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.

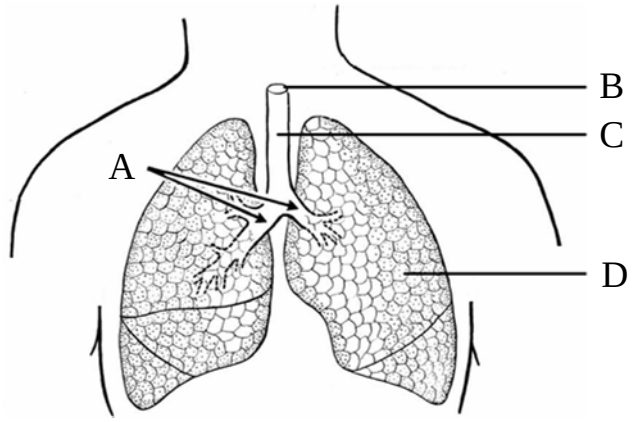


Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

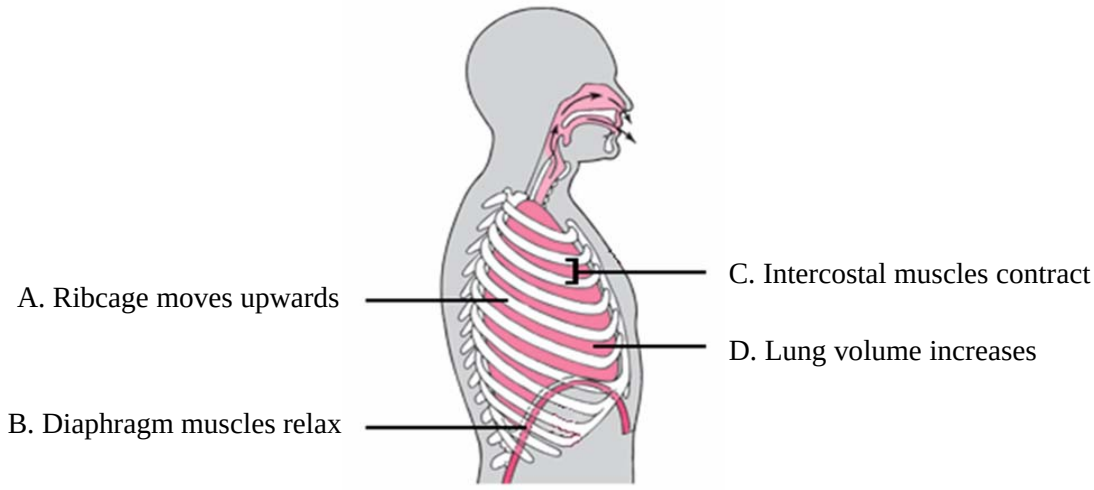
1. The diagram shows the respiratory system of a human being.

   Which of the labelled parts contains sac like structures?



2. The diagram shows the process of exhalation.

   Which of the following is CORRECT about the changes taking place during exhalation?



3. Which of the following is a common characteristic of hydrophytes and halophytes?

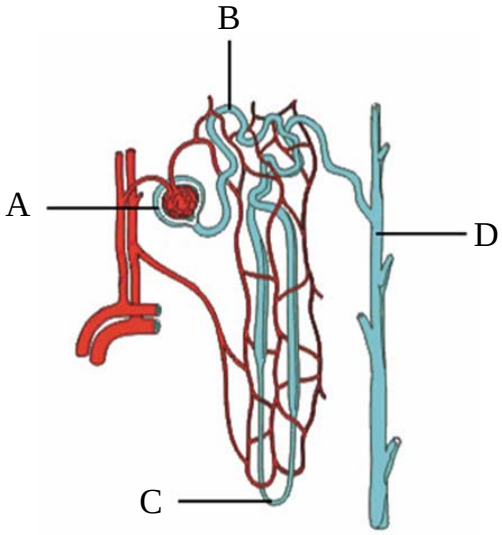
   A. Broad leaves
   B. Deep roots
   C. Water living
   D. Waxy cuticle

4. On a cold day, what happens to erector muscles and hair on the skin to prevent loss of heat from the body?

	Erector Muscles in the Dermis	Hair on the Skin
A.	Contract	Goose bumps
B.	Relax	Goose bumps
C.	Contract	Normal
D.	Relax	Normal

5. The diagram shows the structure of a nephron.

At which of the labelled parts of the nephron does ultrafiltration take place?



6. A person is suffering with loss of appetite and sleeplessness.

Which of the following parts of his brain seems to be affected?

A. Cerebrum
B. Cerebellum
C. Hypothalamus
D. Medulla oblongata

7. The blind spot in an eye is a point where

A. the highest density of cone cells is found.
B. refraction of light rays takes place.
C. a jelly like substance is present.
D. the optic nerve leaves the eye.

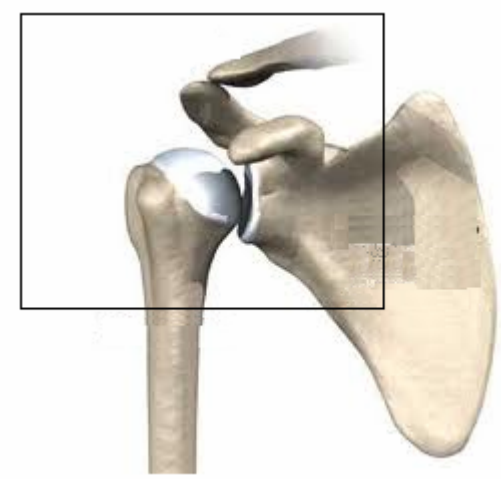
8. An excessive production of thyroxin in humans may result in

A. enlarged internal organs.
B. increased energy production.
C. high blood calcium level.
D. more reabsorption of water by kidneys.

9. All of the following are correct about a ‘bone’ EXCEPT that

A. it is an example of muscular tissue.
B. it stores calcium and phosphate.
C. it produces red and white blood cells.
D. it is the hardest substance in the body.

10. The diagram shows a type of joint found in man.



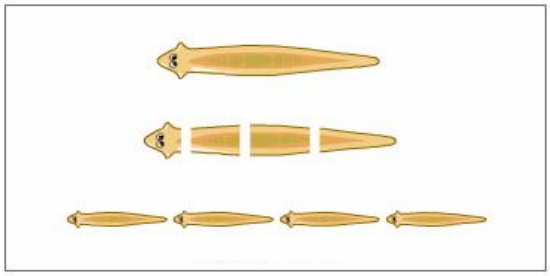
This type of joint is found

A. between the skull bones.
B. between the vertebrae.
C. at the shoulder.
D. at the elbow.

11. The effects of arthritis are more visible at the

A. neck joint.
B. elbow joint.
C. wrist joint.
D. ankle joint.

12. Which of the following methods of asexual reproduction is shown in the given diagram?



A. Spore formation
B. Binary fission
C. Fragmentation
D. Budding

13. In flowering plants, the fusion of male and female gametes results in the formation of a/an

A. zygote.
B. embryo.
C. endosperm.
D. pollen tube.

14. In rabbits, the secretions released by the seminal vesicles help to

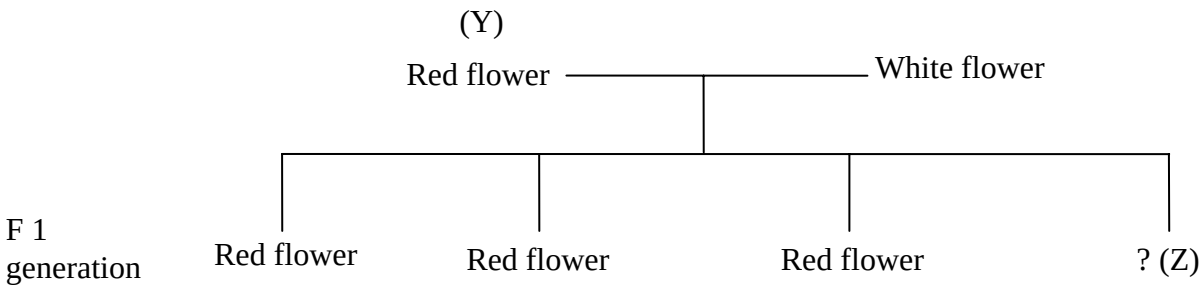
A. form sperms inside the testes.
B. provide nutrients to the sperm cell.
C. neutralize the semen.
D. prevent sexually transmitted diseases.

15. In a particular variety of cats, the fur colour is controlled by B and b. ('B' denotes black and 'b' denotes white)

Which of the following crosses would result in a 1:1 ratio of the two fur colours in the F 1 generation?

A. BB × bb
B. Bb × bb
C. Bb × BB
D. Bb × Bb

16. The given pedigree shows the resultant of a cross between red and white flowers.



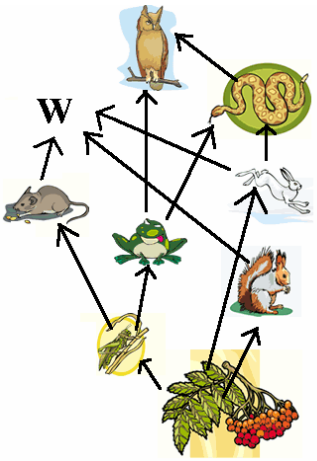
Which of the following CORRECTLY identifies Y and Z?

	Genotype of Red Parent Flower (Y)	**Missing Phenotype (Z)**
A.	RR	Red
B.	Rr	Red
C.	RR	White
D.	Rr	White

17. Which of the following pairs is an example of discontinuous variation?

A. Blood group and height
B. Height and gender
C. Gender and tongue rolling
D. Tongue rolling and intelligence

18. The diagram shows a food web.



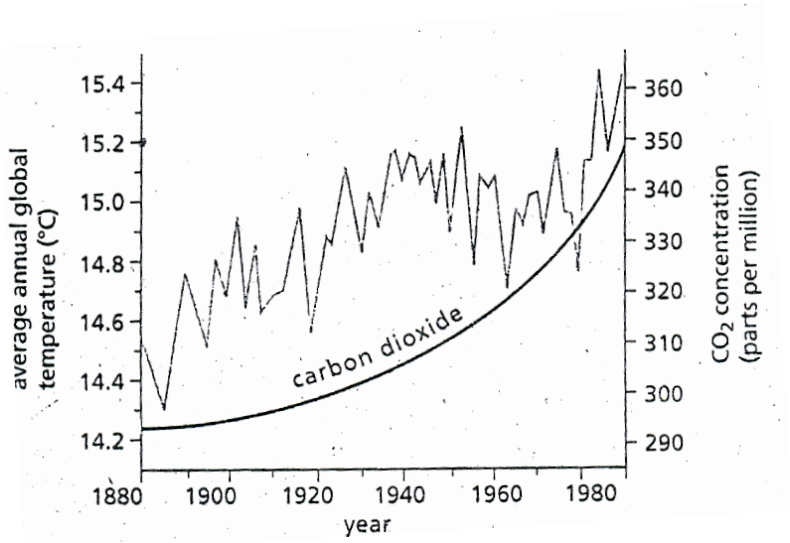
The missing living organism 'W' could be a

A. fox.
B. horse.
C. lizard.
D. caterpillar.

19. The role of nitrogen-fixing bacteria is to convert atmospheric nitrogen first into

A. nitrites.
B. nitrates.
C. plant proteins.
D. ammonium compounds.

20. The graph shows annual average global temperatures till year 1980.



Which of the following environmental phenomenon is focused in this graph?

A. Acid rain
B. Green house effect
C. Cutting down of trees
D. Urbanization

21. A predator is an organism which

A. lives inside a living host.
B. captures sunlight to make its food.
C. attacks and eats another organism.
D. decomposes dead bodies to get energy.

22. 'The bacteria act on the sugar, lactose, and convert it to lactic acid.'

The above statement is one of the steps in the production of

A. yoghurt.
B. bread.
C. alcohol.
D. antibiotics.

23. In an industrial fermenter, an impeller is used to

A. remove excess heat from the system.
B. mix the air bubbles with the nutrient broth.
C. keep an acid-base balance of the medium.
D. prevent the entry of unwanted microorganisms.

24. Cephalosporins and tetracyclines are the examples of

A. vaccines.
B. sedatives.
C. antibiotics.
D. pain killers.

25. The common feature among sedatives, narcotics and hallucinogens is that they all

A. are derived from *Cannabis sativa*.
B. interact with nervous system of the body.
C. are non-addictive and easily quitable.
D. relieve pain and are given to cancer patients.

END OF PAPER

Please use this page for rough work